Polarization of the CMB: Deciphering E- & B-Modes

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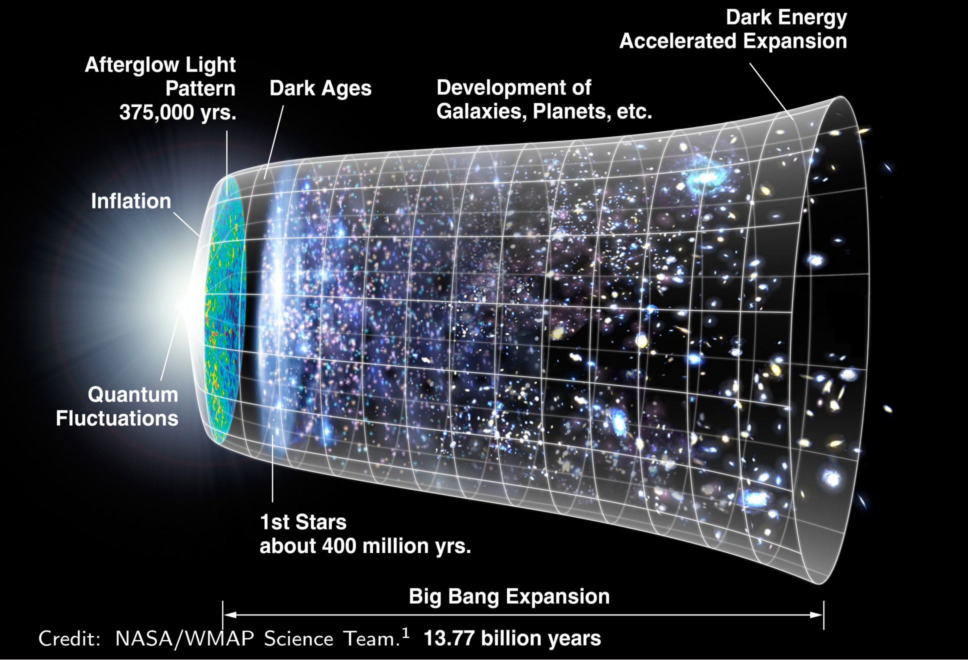
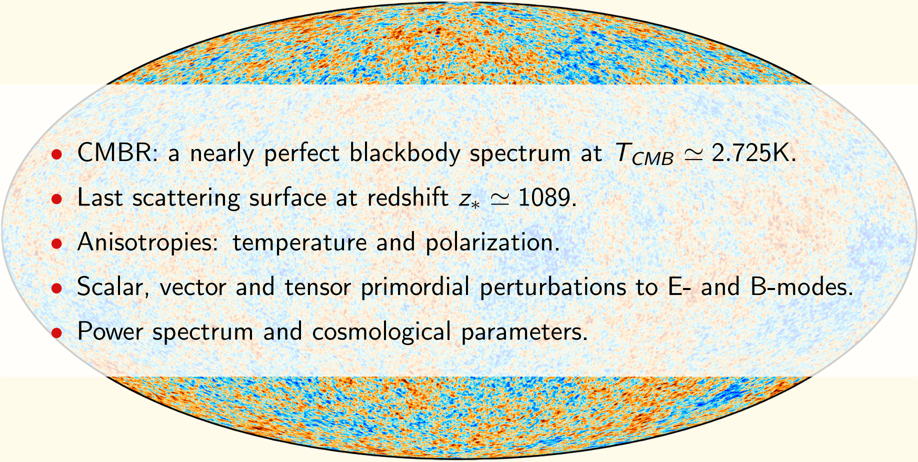
Outline

1. Introduction i



Credit: NASA/WMAP Science Team.[1]

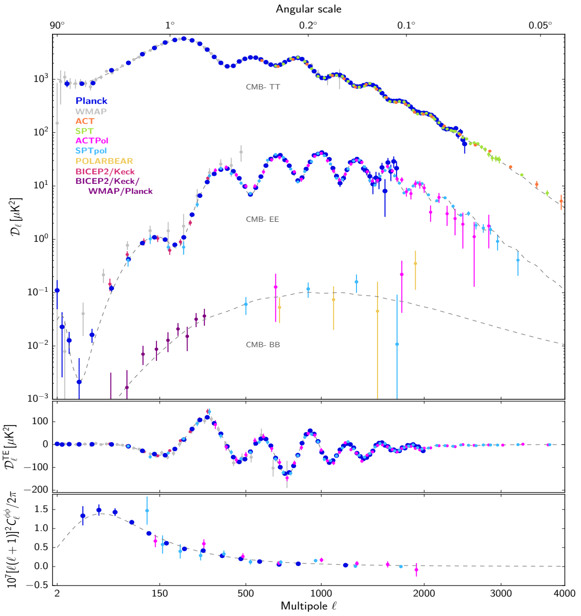
1. Introduction ii



- CMBR: a nearly perfect blackbody spectrum at $T_{CMB} \simeq 2.725$K.
- Last scattering surface at redshift $z_* \simeq 1089$.
- Anisotropies: temperature and polarization.
- Scalar, vector and tensor primordial perturbations to E- and B-modes.
- Power spectrum and cosmological parameters.

Credit: ESA and the *Planck* Collaboration.[2]

1. Introduction iii



Credit: Wikipedia contributors, "Cosmic microwave background — Polarization".[3]

- The polarization of the CMBR can be described by the **Stokes' parameters** I, Q and U, which are defined by the following matrix

$$I_{ij} = \begin{pmatrix} I+Q & U \\ U & I-Q \end{pmatrix}, \tag{1}$$

where I represents total intensity, Q and U define linear polarization.

- There is no circular polarization V.

Light	**Stokes' parameters conditions**
Unpolarized	$Q = 0, \quad U = 0$
Linearly Polarized	$Q \neq 0 \quad \vee \quad U \neq 0$

Table 1: Stokes parameters for polarization states.

- The **flat-sky approximation** is used to describe the polarization field, which is valid for small angular scales.

- The **Fourier transform** of the polarization field can be defined as:

$$P(\ell) = \int d^2\theta \, P(\theta) e^{i\ell \cdot \theta}, \tag{2}$$

where where ℓ is the multipole moment, which is the Fourier conjugate of the angular position, θ, on the sky.

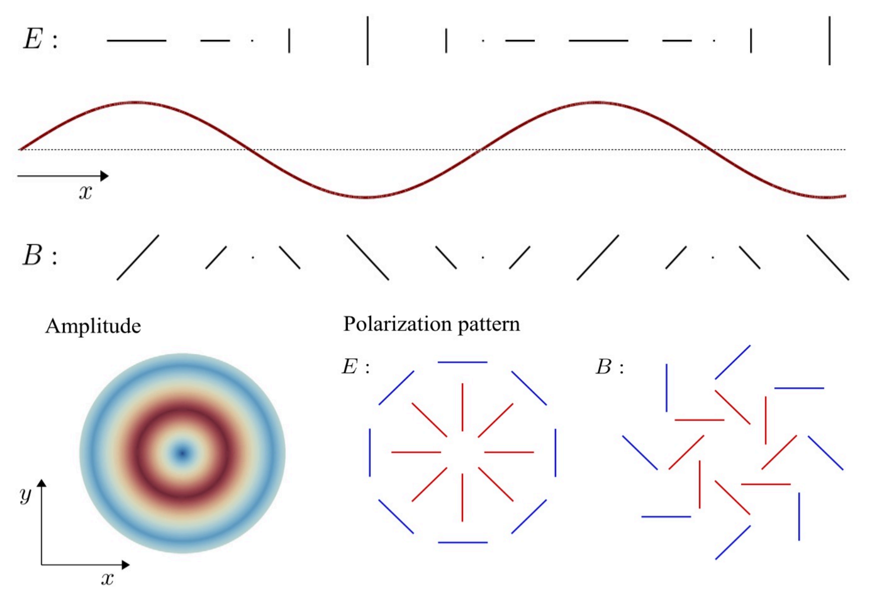
- Defining the **azimuthal angle**, ϕ_ℓ, of the 2D wavevector, $\boldsymbol{\ell} = (\ell_x, \ell_y) = \ell_0(\cos\phi_\ell, \sin\phi_\ell)$, the E- and B-mode become:

$$E(\ell) = \cos(2\phi_\ell)Q(\ell) + \sin(2\phi_\ell)U(\ell) \tag{3}$$

$$B(\ell) = -\sin(2\phi_\ell)Q(\ell) + \cos(2\phi_\ell)U(\ell) \tag{4}$$

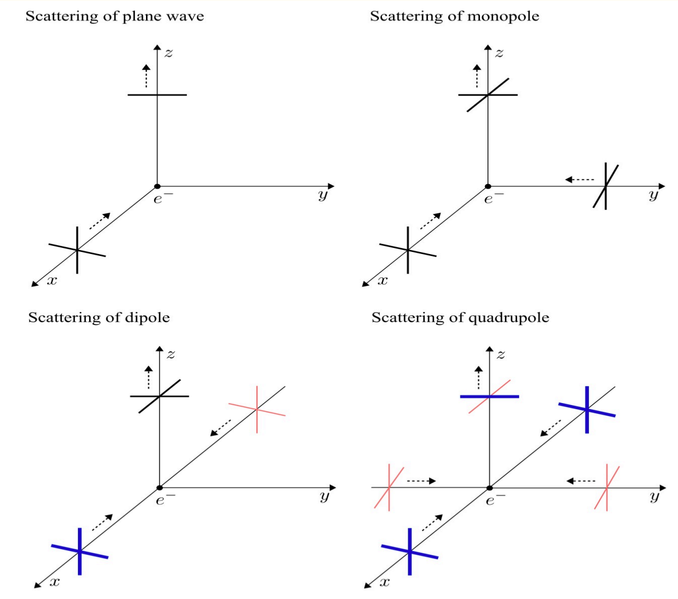
Property	E- (gradient-) modes	B- (curl-) modes
Mathematical nature	Curl-free ($\nabla \cdot \mathrm{B} = 0$) Scalar: $E(\ell)$	Divergence-free ($\nabla \cdot \mathrm{E} = 0$) Pseudo-scalar: $B(\ell)$
Symmetry	Even parity	Odd parity
Primordial source	Scalar perturbations	Tensor perturbations
Signal strength	Dominant polarization	$\sim$ One order of magnitude weaker

Table 2: Physical properties and primordial origins of CMB (E- and B-) polarization modes.



Credit: S. Dodelson and F. Schmidt, *Modern Cosmology*, 2nd ed. Academic Press, 2020.[4]

2. Physics of CMB Polarization v



Credit: S. Dodelson and F. Schmidt, *Modern Cosmology*, 2nd ed. Academic Press, 2020.[4]

- **Scalar or compressional perturbations** are energy density perturbations in the early Universe.
- **Dominant** source of temperature anisotropies and E- polarization in the CMB.
- **Do not** generate B-mode polarization due to their even parity.
- This type of perturbation generates quadrupole anisotropies with $(l, m) = (2, 0)$, which are represented by the following spherical harmonic:

$$Y_2^0(\theta, \phi) = \sqrt{\frac{5}{16\pi}}(3\cos^2\theta - 1), \tag{5}$$

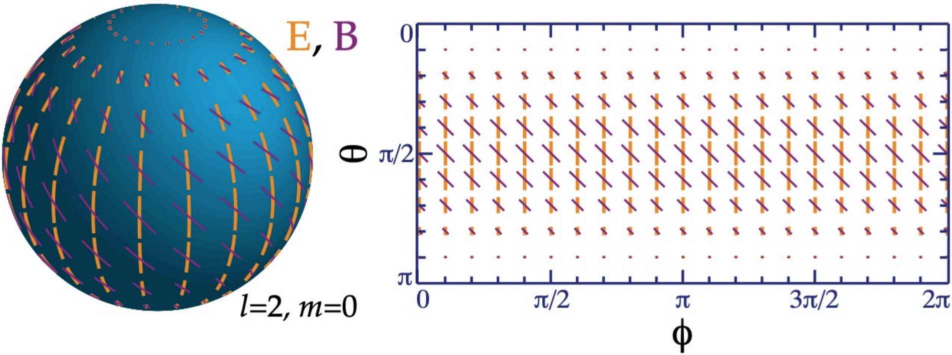
- and whose polarization pattern can be described as a "*pure Q-field*" with no U component,

$$Q = \sin^2\theta, \quad U = 0, \tag{6}$$

- This pattern generates a polarization that is parallel or perpendicular to the z-axis, which corresponds to the E-mode polarization, shown in the next slide.

$$Y_2^0(\theta,\phi) = \sqrt{\frac{5}{16\pi}}(3\cos^2\theta - 1), \quad (5)$$

$$Q = \sin^2\theta, \quad U = 0, \quad (6)$$



Credit: W. Hu and M. White, "A CMB polarization primer," *New Astronomy*, vol. 2, no. 4, pp. 323–344, 1997.[5]

- **Vector or vortical perturbations** are perturbations in the velocity field of the primordial plasma.
- They can generate **both** E-mode and B-mode polarization patterns.
- However, they **decay rapidly** in the early Universe, so their contribution to the polarization is **negligible** compared to the scalar and tensor perturbations as their amplitude decreases faster.
- This type of perturbation generates quadrupole anisotropies with $(l, m) = (2, 1)$, which are represented by the following spherical harmonic:

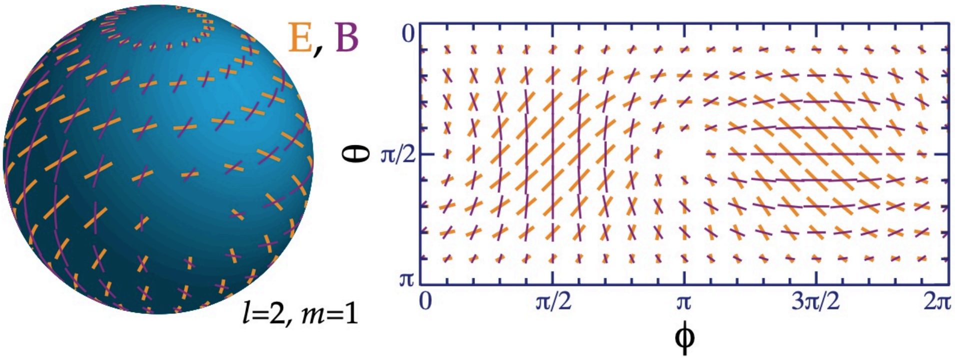
$$Y_2^{\pm 1}(\theta, \phi) = \sqrt{\frac{15}{8\pi}} \sin\theta \cos\theta e^{\pm i\phi}. \quad (7)$$

- The polarization pattern generated by the vector perturbations is the following:

$$Q = -\sin\theta \cos\theta e^{i\phi}, \quad U = -i \sin\theta e^{i\phi}, \quad (8)$$

- This pattern is shown in the next slide.

$$Y_2^{\pm 1}(\theta,\phi) = \sqrt{\frac{15}{8\pi}} \sin\theta\cos\theta e^{\pm i\phi} \qquad (7) \qquad | \qquad Q = -\sin\theta\cos\theta e^{i\phi}, \quad U = -i\sin\theta e^{i\phi} \quad (8)$$



Credit: W. Hu and M. White, "A CMB polarization primer," *New Astronomy*, vol. 2, no. 4, pp. 323–344, 1997.[5]

2.3. Tensor perturbations i

- **Tensor or gravity waves perturbations** are perturbations in the metric of spacetime itself.
- They can generate **both** E-mode and B-mode polarization patterns.
- However, the amplitude is expected to be an order of magnitude weaker than that of the scalar perturbations, so their contribution to the CMB polarization is small.
- This type of perturbation generates quadrupole anisotropies with $(l, m) = (2, 2)$, which are represented by the following spherical harmonic:

$$Y_2^{\pm 2}(\theta, \phi) = \sqrt{\frac{15}{32\pi}} \sin^2 \theta e^{\pm 2i\phi}, \tag{9}$$

- The polarization pattern generated by the tensor perturbations is the following:

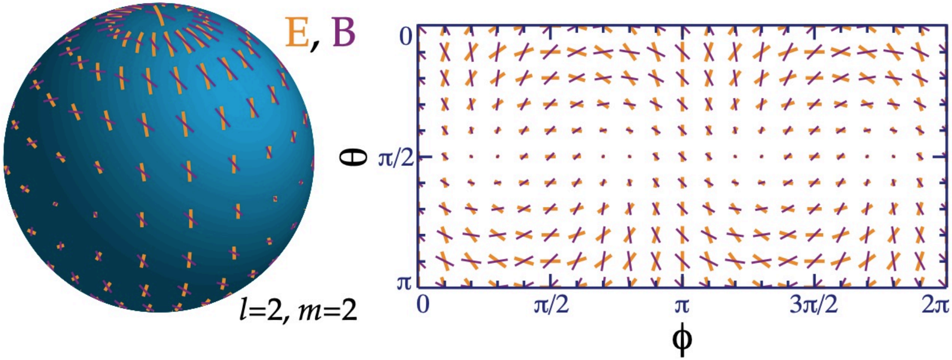
$$Q = (1 + \cos^2 \theta) e^{2i\phi}, \quad U = -2i \cos \theta e^{2i\phi}, \tag{10}$$

- This pattern is shown in the next slide.

2.3. Tensor perturbations ii

$$Y_2^{\pm 2}(\theta,\phi) = \sqrt{\frac{15}{32\pi}}\sin^2\theta e^{\pm 2i\phi} \quad (9)$$

$$Q = (1+\cos^2\theta)e^{2i\phi}, \quad U = -2i\cos\theta e^{2i\phi} \quad (10)$$



Credit: W. Hu and M. White, "A CMB polarization primer," *New Astronomy*, vol. 2, no. 4, pp. 323–344, 1997.[5]

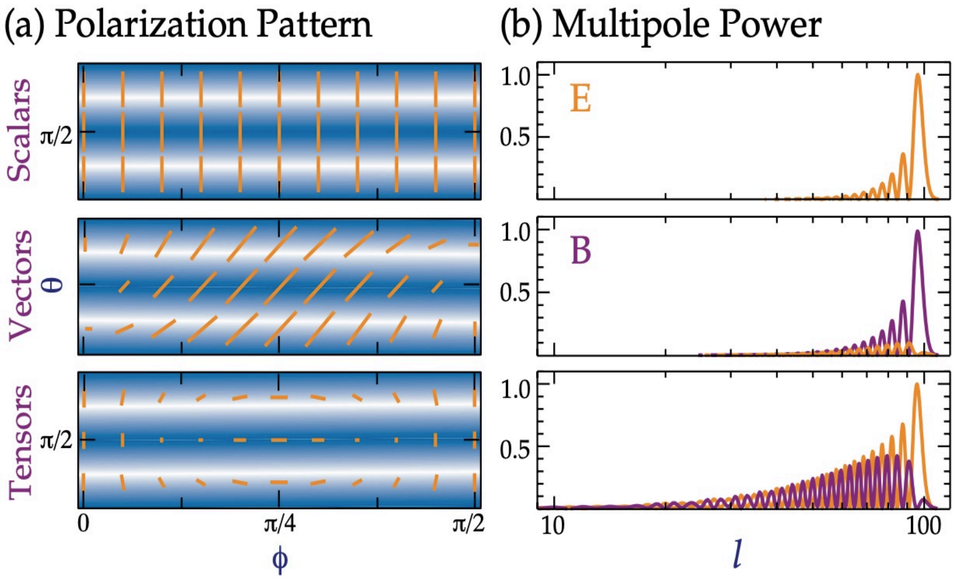
2.4. Summary of perturbations and polarization patterns i

- **Scalar** perturbations generate E-modes with a pure Q-field pattern, while **vector** and **tensor** create both E- and B-modes with mixed patterns. However, vector perturbations decay early in the Universe and tensor perturbations are an order of magnitude weaker than scalar perturbations.

- Certain polarization patterns keep their parity and correlation with temperature fluctuations even after the **superposition** of the different perturbations in the early Universe.

Perturbation Type	Scalar	Vector	Tensor
Amplitude Ratio (B/E)	0	6	8/13

Table 3: Amplitude ratios (B/E) across primordial perturbation modes.



Credit: S. Dodelson and F. Schmidt, *Modern Cosmology*, 2nd ed. Academic Press, 2020.[4]

E-modes

- Mainly generated by **scalar perturbations**, which are the dominant source of CMB anisotropies.
- However, they **can** also be generated by vector and tensor perturbations, but their contribution is smaller.
- They are curl-free and have even parity, and their polarization pattern parallel, $\|$, or perpendicular, $\perp$, to the wavevector, ℓ, in Fourier space.

- In order to calculate the power spectrum for the E-modes, which is on of the most useful tools in cosmology, the expression for the **polarization moments**

$$E_\ell(k) \simeq -\frac{5k}{6\tau'(\eta_*)}\Theta_1(k,\eta_*)\frac{\ell^2}{(k\eta_0)^2}j_\ell(k\eta_0), \tag{11}$$

Symbol	Physical Definition
Θ_1	Dipole moment of the temperature anisotropies
τ'	Optical depth at recombination (backward in time)
k	Wavenumber of the perturbation
η_*	Conformal time at recombination
η_0	Conformal time today
j_ℓ	Spherical Bessel function of order ℓ

Assumptions:[4] *i*) tight-coupling regime between photons and e^- prior to recombination, *ii*) polarization condition $\Theta_P = \Theta_E$.

Table 4: Mathematical variables inside the polarization anisotropy expressions.

3. E-modes iii

- The **transfer function** for the E-modes is defined in the same way as for the temperature anisotropies, as the ratio of the moments, $E_\ell(k)$, to the primordial curvature perturbation, $\mathcal{R}(\mathbf{k})$:

$$\mathcal{T}_\ell^E(k) \equiv \frac{E_\ell(k)}{\mathcal{R}(\mathbf{k})}, \tag{12}$$

- The **power spectrum of the E-modes,** C_{EE}, is the Fourier transform of the two-point (auto-)correlation function of the E-mode polarization field.

$$C_{EE}(\ell) = \frac{2}{\pi}\int_0^\infty k^2 dk \left|\mathcal{T}_\ell^E(k)\right|^2 P_{\mathcal{R}}(k), \tag{13}$$

where $P_{\mathcal{R}}(k)$ is the power spectrum of the primordial curvature perturbation.

3. E-modes iv

- The same can be done for the temperature anisotropies,

$$C_{TT}(\ell) = \frac{2}{\pi}\int_0^\infty k^2 dk \left|\mathcal{T}_\ell^{T}(k)\right|^2 P_{\mathcal{R}}(k). \tag{14}$$

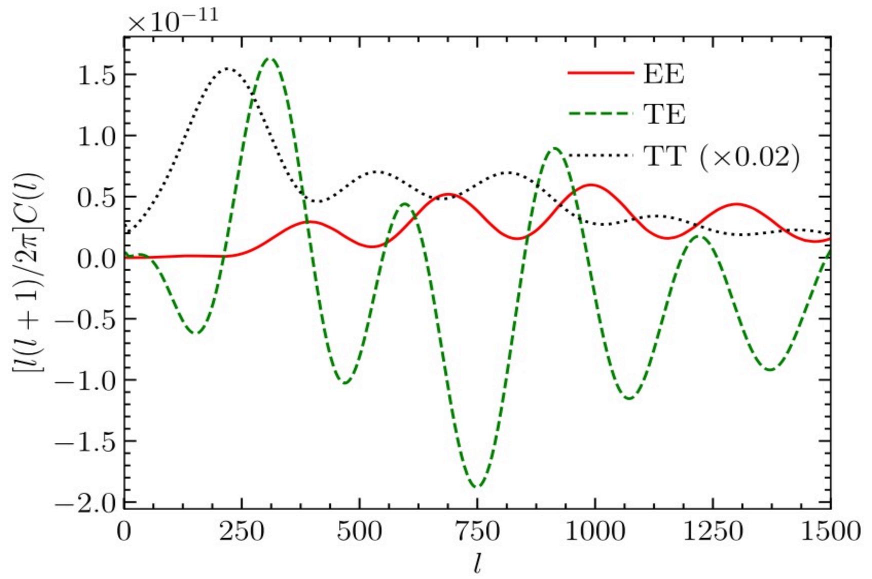
- Finally, the **cross-correlation power spectrum** between the temperature and E-mode polarization, C_{TE},

$$C_{TE}(\ell) = \frac{2}{\pi}\int_0^\infty k^2 dk \left|\mathcal{T}_\ell^{T^*}(k)\mathcal{T}_\ell^{E}(k)\right| P_{\mathcal{R}}(k). \tag{15}$$

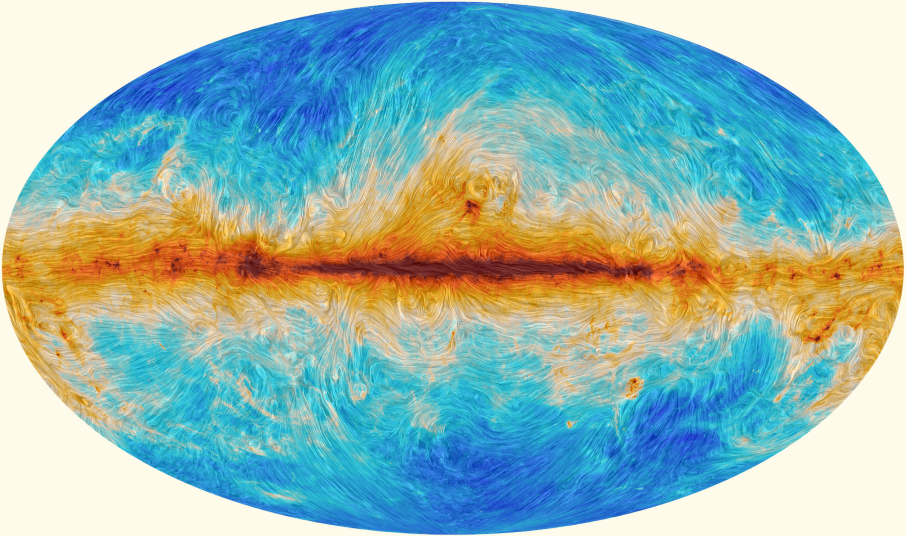
- However, the standard convention to plot the cross-correlation power spectrum (and many other power spectra) is defined as

$$\mathcal{D}_\ell^{TE} \equiv \frac{\ell(\ell+1)}{2\pi} C_{TE}(\ell), \tag{16}$$

where the factor $\ell(\ell+1)/2\pi$ is introduced to make the power spectrum more flat and easier to visualize.



Credit: S. Dodelson and F. Schmidt, *Modern Cosmology*, 2nd ed. Academic Press, 2020.[4]

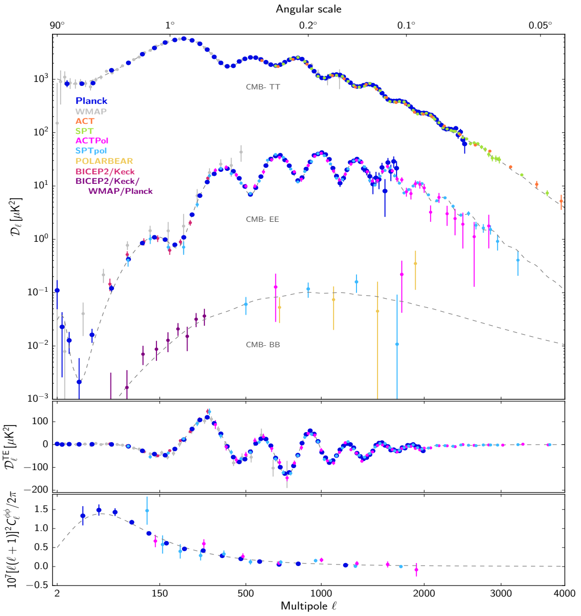


Credit: ESA and the *Planck* Collaboration.[2]

B-modes

- Primordially generated by **vector** and **tensor perturbations**.
- **Not** generated at all by scalar perturbations, then $C_{BB}^{\text{scalar}} = 0$.
- Can be **contaminated** by *i*) **foreground emission** from our galaxy and *ii*) **gravitational lensing** of E-modes, which can produce a **modulation** of the E-mode polarization pattern and create a B-mode pattern.
- They are divergence-free and have odd parity, and their polarization pattern is at 45° with respect to the wavevector, ℓ, in Fourier space.
- Expected to have **about a tenth** of the power of the E-mode, so they are much more challenging to detect.

4. B-modes ii



Credit: Wikipedia contributors, "Cosmic microwave background — Polarization".[3]

4. B-modes iii

- Perturbations outside the horizon at recombination so that $C_{TE} \neq 0$ on large scales ($\ell \lesssim 100$). In agreement with inflationary theory: initial perturbations generated before recombination, during inflationary epoch.

- **No cross-correlation** for temperature and magnetic polarization ($C_{TB} = 0$) nor for electric and magnetic polarization ($C_{EB} = 0$) due to **opposite parity**.

- **Little data** has been obtained for the B-mode power spectrum, **but** it seems to be in agreement with the predictions of the ΛCDM model.

- **Lensing deflection** must be subtracted to detect the primordial B-modes generated by tensor perturbations.

- **Diffusion damping** causes all power spectra to decay at large ℓ (small scales) because of the finite thickness of the last scattering surface.

Current (and past) Observations

- **COBE:** Satellite. NASA. 1989-1993. First to measure the CMBR temperature anisotropies and to determine that CMBR is close to a blackbody.[7]

- **BOOMERanG:** Balloon-borne. International collab. 1997. First high resolution measurements of CMB temperature anisotropies. Evidence of a flat Universe.[8]

- **DASI:** Ground-based. South Pole. 2000-2008. First to detect the 2nd and 3rd acoustic peaks in the CMB temperature p.s. 2002: first detection of E-mode polarization of the CMB. Also measured TE cross-correlation p.s.[9]

- **WMAP:** Satellite. NASA. 2001-2010. Improved COBE satellite sensitivity and angular resolution. Measured $H_0 = 69.32 \pm 0.80$ km/s/Mpc.[10]

5. Current Observations and Future Prospects ii

- **BICEP/Keck Array:** Ground-based. South Pole. 2006-present. They intend to measure the B-mode polarization of the CMB. 2014: BICEP2 detected B-mode polarization, but was a contamination from the galactic dust.[11]

- **Planck:** Satellite. ESA. 2009-2013. Improved WMAP satellite sensitivity and angular resolution. Better CMB temperature anisotropies and polarization measurements than WMAP.[12]

- **POLARBEAR:** Ground-based. International collab. at Atacama Desert. 2012-present. Discovered the B-mode polarization by E-modes gravitational lensing in 2014. Deeper field than BICEP2, dust contamination smaller.[13]

- **Simons Observatory:** Ground-based. Simons Foundation at Atacama Desert. 2024-present. Expected to improve r (tensor-to-scalar ratio measurement) sensitivity by an order of magnitude with respect to Planck satellite.[14]

Future Prospects

- **LiteBIRD:** Satellite. JAXA for 2032. Goal: measure the B-mode polarization of CMB at multifrequency to separate the primordial B-modes from foreground contamination.[15]

- **PICO:** Satellite. NASA. No launch date. 21 frequency channels and a large number of detectors, then high sensitivity to the CMB polarization.[16]

- **PIPER:** Balloon-borne. NASA. No launch date. Meant to measure the B-mode polarization up to three times smaller than the lowest predicted value.[17]

6. Conclusions i

1. Photons get polarized by **Thomson scattering** because of scalar, vector and tensor perturbations in the early Universe.
2. E- and B-mode polarization patterns have **different generation mechanisms and parity properties**, which allows to disentangle the contributions to the CMB polarization.
3. E-mode polarization is well determined, B-mode polarization not yet.
4. No B-mode polarization is generated by scalar perturbations The detection of primordial B-mode polarization implies the existance of tensor perturbations, or to say, **primordial gravitational waves**, predicted by inflationary theory.
5. Future experiments are focused on the detection of this B-mode polarization.

7. References

1. https://science.nasa.gov/mission/wmap/wmap-overview/#mission
2. https://www.cosmos.esa.int/web/planck/picture-gallery
3. https://en.wikipedia.org/wiki/Cosmic_microwave_background#Polarization
4. S. Dodelson and F. Schmidt, *Modern Cosmology*, 2nd ed. Academic Press, 2020.
5. https://arxiv.org/abs/astro-ph/9706147
6. https://arxiv.org/abs/1807.06205

Other Wikipedia contributors:

7. "Cosmic Background Explorer," https://en.wikipedia.org/wiki/Cosmic_Background_Explorer
8. "BOOMERanG experiment," https://en.wikipedia.org/wiki/BOOMERanG_experiment
9. "Degree Angular Scale Interferometer," https://en.wikipedia.org/wiki/Degree_Angular_Scale_Interferometer
10. "Wilkinson Microwave Anisotropy Probe," https://en.wikipedia.org/wiki/Wilkinson_Microwave_Anisotropy_Probe
11. "BICEP and Keck Array," https://en.wikipedia.org/wiki/BICEP_and_Keck_Array
12. "Planck (spacecraft)," https://en.wikipedia.org/wiki/Planck_(spacecraft)
13. "POLARBEAR," https://en.wikipedia.org/wiki/POLARBEAR
14. "Simons Observatory," https://en.wikipedia.org/wiki/Simons_Observatory
15. "LiteBIRD," https://en.wikipedia.org/wiki/LiteBIRD

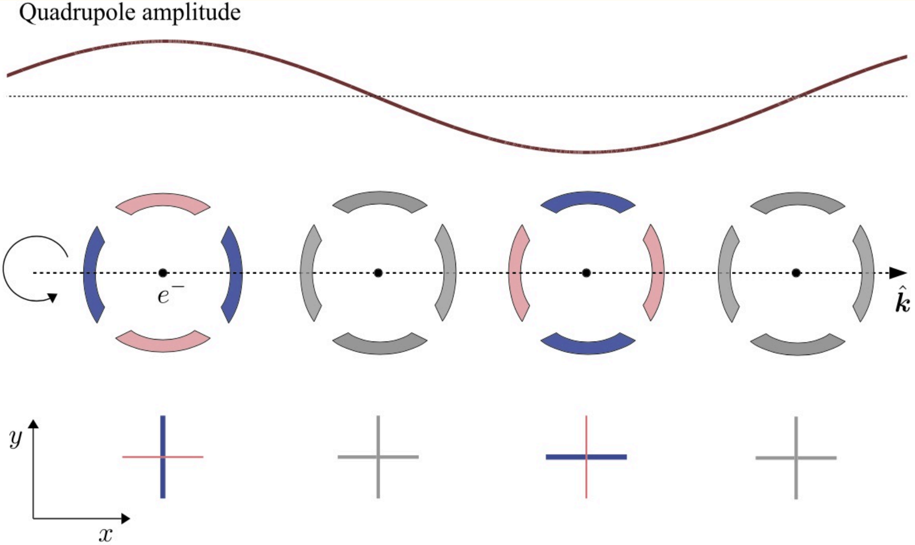
NASA:

16. "PICO: Probe of Inflation and Cosmic Origins," https://sites.google.com/umn.edu/picomission
17. "PIPER: Primordial Inflation Polarization ExploreR," https://asd.gsfc.nasa.gov/piper/

Thanks for your attention!

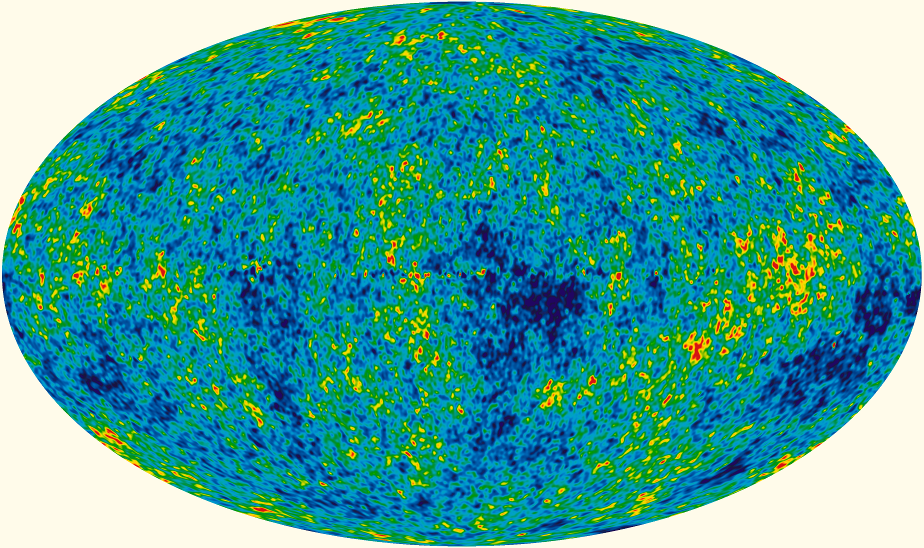
Round of questions.

E-mode: scalar perturbation in $\hat{e}_x$



Credit: S. Dodelson and F. Schmidt, *Modern Cosmology*, 2nd ed. Academic Press, 2020.[4]

WMAP



Credit: ESA and the *Planck* Collaboration.[2]